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Assessment Cover Page

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Module Title	Data Visualization Techniques, Machine Learning for Business
Assessment Title	Integrated CA2_DVT_MLB
Assessment Due Date	21/05/2025
Date of Submission	

Declaration

By submitting this assessment, I confirm that I have read the CCT policy on academic misconduct and understand the implications of submitting work that is not my own or does not appropriately reference material taken from a third party or other source.

I declare it to be my own work and that all material from third parties has been appropriately referenced.

I further confirm that this work has not previously been submitted for assessment by myself or someone else in CCT College Dublin or any other higher education institution.

1. Introduction

In the contemporary digital economy, e-commerce platforms produce substantial amounts of client data that can be utilised to enhance user experience, simplify operations, and boost sales. Online merchants encounter the issue of interpreting data to facilitate informed decision-making, which necessitates the use of data visualisation and machine learning (ML) tools.

According to Provost and Fawcett (2013), data science allows companies to derive important insights from data and convert them into effective strategies, particularly in dynamic and competitive sectors like e-commerce.

This project aims to analyse transactional and customer-level data from the Brazilian e-commerce platform Olist to derive business insights via visual storytelling and machine learning models. The evaluation is organised into three fundamental tasks:

1. The creation and assessment of a recommendation system employing collaborative filtering methodologies.
2. A Market Basket Analysis (MBA) employing the Apriori and FP-Growth algorithms to discern relationships among items.
3. The development of an interactive dashboard designed for young adults (ages 18–35), emphasising significant data trends via dynamic visualisations.

2. Recommendation Systems

Recommendation Systems are integral to online retail platforms, enhancing user experience, augmenting engagement, and elevating sales. Recommendation systems can forecast pertinent goods for each user by examining customer behaviour, preferences, and purchase history, which improves satisfaction and loyalty. In competitive markets such as e-commerce, personalisation serves as a crucial differentiation, with recommendation engines being a fundamental facilitator of this approach (Ricci et al., 2015).

There are two primary categories of recommendation systems: content-based filtering and collaborative filtering. Content-based filtering utilises item characteristics and user profiles to generate recommendations. For instance, if a user acquires noise-cancelling headphones, the system may suggest more audio-related products based on factors such as category, brand, or price. Conversely, collaborative filtering emphasises user behaviour above product attributes. It suggests things based on the preferences or purchases of analogous users, irrespective of the item's substance.

This project employed collaborative filtering through two methodologies:

- **User-user filtering** detects analogous users based on purchasing behaviours and suggests things that those comparable users have acquired.
- **Item-item filtering** identifies things that are commonly purchased in conjunction and recommends analogous items based on their co-occurrence.

The technique utilised a user-product matrix derived from purchase data, then employing cosine similarity to assess associations between people and goods. Owing to data sparsity and the characteristics of the dataset, item-item filtering provided more significant and consistent outcomes. For instance, consumers who acquired products in the "audio" category were often linked to purchases in "watches_gifts" and "computers_accessories," indicating potential for bundled recommendations.

Collaborative filtering was selected instead of content-based approaches due to the dataset's deficiency in comprehensive product metadata (e.g., brand, technical specifications) essential for an effective content-based model. Also, collaborative filtering utilises genuine user behaviour, enhancing the personalisation and relevance of recommendations.

These recommendation models illustrate the capacity of machine learning to raise conversion rates and improve consumer engagement for merchants. Deploying such solutions on an e-commerce platform can facilitate upselling methods, enhance product discovery, and augment average basket size.

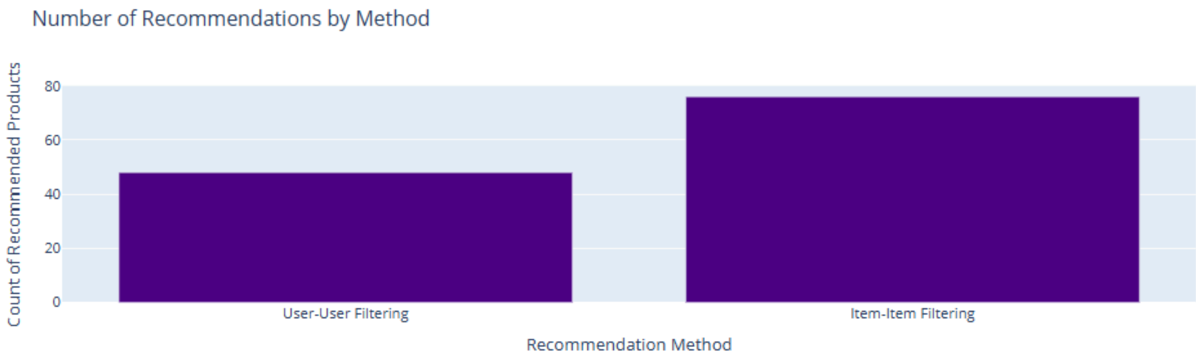

Figure 1 - Number of recommendations generated by each collaborative filtering method.

Figure 1 contrasts the quantity of product recommendations produced by the user-user and item-item collaborative filtering techniques. Item-item filtering yielded a greater quantity of pertinent recommendations, signifying enhanced scalability and performance for this dataset. This substantiates the decision to prioritise item-based models for future implementation in e-commerce contexts.

3. Market Basket Analysis

Market Basket Analysis (MBA) is a data mining methodology employed to reveal connections and co-occurrence patterns among products in transactional datasets. In the realm of online retail, it enables businesses to identify products that are commonly purchased in conjunction, consequently facilitating cross-selling, upselling, and product placement techniques (Tan et al., 2005).

The MBA was conducted utilising the Apriori and FP-Growth algorithms from the **mlxtend library**. The information was initially compiled by integrating order, product, and translation data to achieve an organised transactional perspective featuring product category names in

English. The dataset was subsequently converted into a binary matrix, with each row denoting a transaction (*order_id*) and each column signifying a product category.

A primary problem was that most orders in the dataset had only a single item, hindering the identification of significant relationship rules. The information was filtered to encompass only transactions including two or more distinct products, facilitating the creation of valid itemsets.

The Apriori algorithm was initially implemented with a minimum support of 0.005 and a lift threshold of 0.5. This generated 76 robust association rules. The FP-Growth algorithm was again employed with identical parameters, yielding the same set of rules, so affirming the consistency and validity of the results.

Among the most robust connections discovered were:

- {*watches_gifts*} → {*audio*} with a lift of 18.15
- {*stationery*} ↔ {*luggage_accessories*} with a lift of 7.8
- {*bed_bath_table*} ↔ {*home_comfort*} with a lift of 3.15

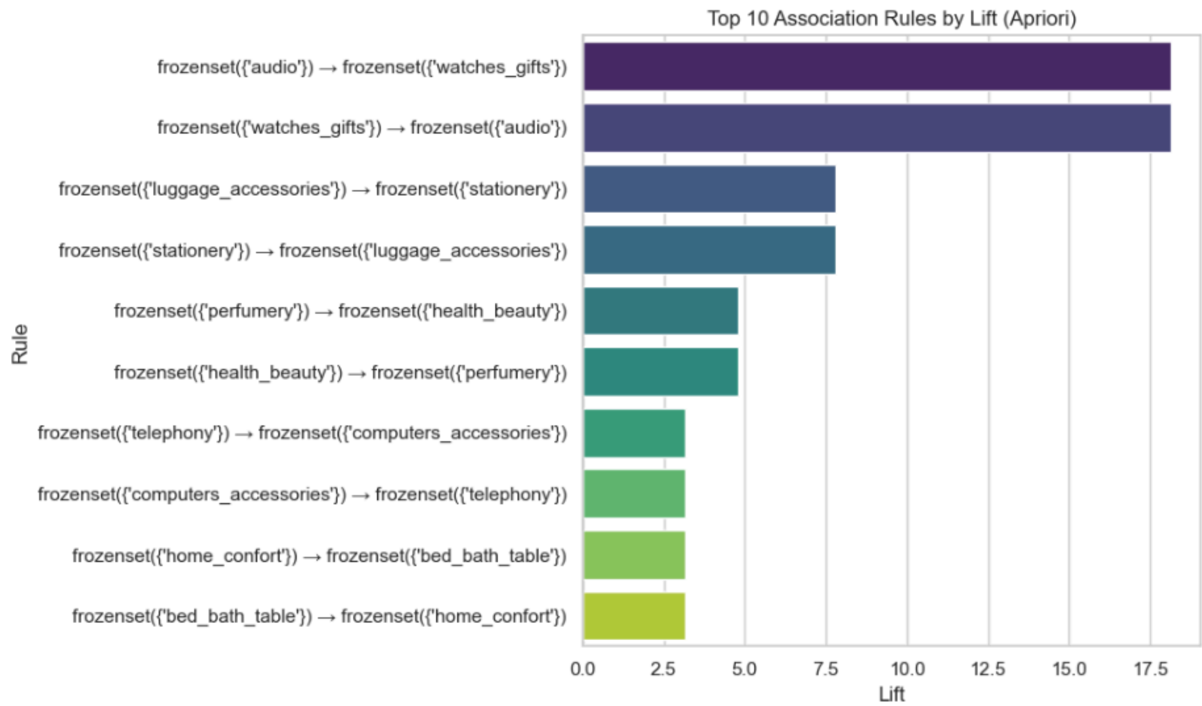

Figure 2 - Top 10 product association rules by lift.

Figure 2 above displays the top 10 product association rules arranged by lift, demonstrating the most robust co-purchase links within the dataset. These regulations embody significant behavioural trends applicable to product bundling and cross-selling efforts.

These rules disclose significant cross-category purchase trends that might be utilised to enhance product recommendations, package marketing, and website design methods.

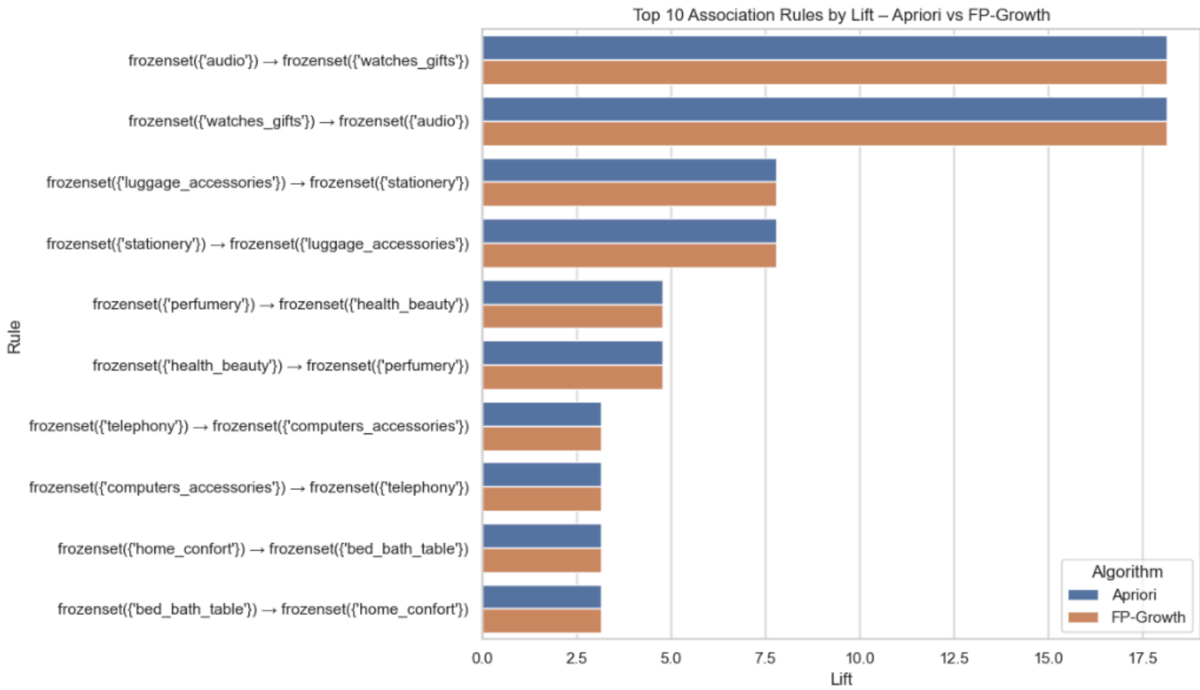

Figure 3 - Apriori vs FP-Growth:

Figure 3 presents an side-by-side of the rules produced by the Apriori and FP-Growth algorithms. The consistent results validate the strength of the relationships and the efficacy of FP-Growth for future scalability.

The MBA methodology enhances the recommendation engine by delivering item-level co-occurrence patterns independent of user profiles. This improves the reliability of product recommendations and facilitates advanced selling methods.

4. Dashboard and Visualisations

An interactive dashboard was created using Plotly in a Jupyter Notebook environment to convey essential business insights clearly and accessibly. The dashboard was specifically developed for younger individuals (ages 18–35), emphasising interactivity, clear visuals, and straightforward navigation—features that connect with the preferences of digitally native users.

The Dashboard includes the following visual components:

1. Top 10 Product Categories

A horizontal bar chart illustrating the most commonly ordered product categories. This graphic elucidates client preferences and assists firms in identifying top-selling product categories for inventory and marketing strategies.

Figure 4 - Top 10 Products Categories by Order Volume.

2. Monthly Order Volume

A time-series line graph displaying the monthly count of unique orders placed. This visualisation illustrates seasonal trends and variations in demand, which can aid in forecasting and campaign planning.

Figure 5 - Monthly Order Volume.

3. Price Distribution by Category

A Plotly graphic that allows users to choose a product category and examine the distribution of product prices. This facilitates the differentiation between premium and budget categories and evaluates pricing methods.

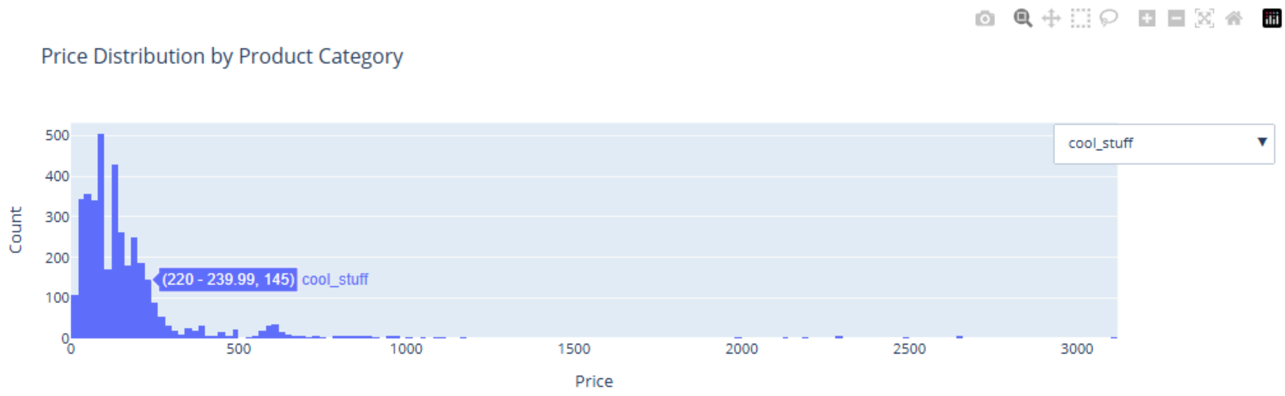

Figure 6 - Price Distribution by Category.

4. Number of Order by Brazilian State

A horizontal bar chart displaying the quantity of orders across many Brazilian states. This chart facilitates area segmentation and logistical planning.

Detailed, category-specific information are beneficial for training machine learning models that classify clients based on price sensitivity or identify pricing anomalies.

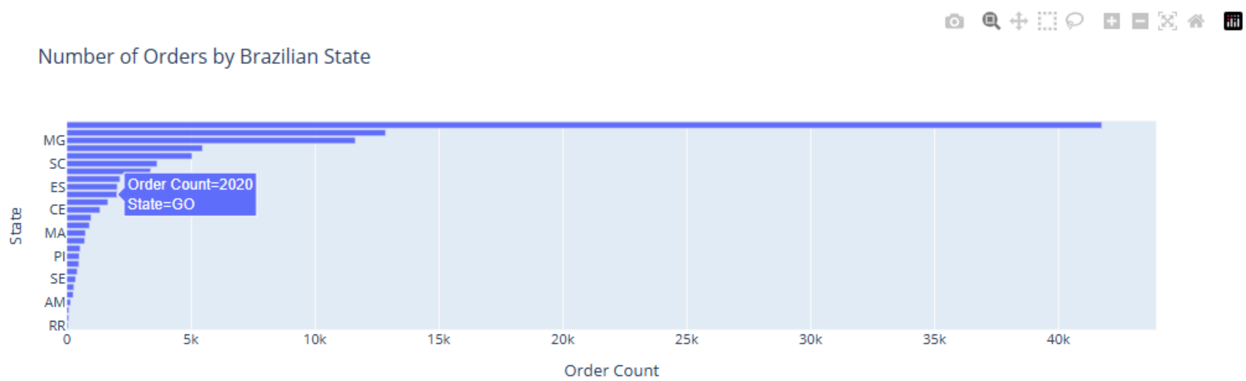

Figure 7 - Orders by Brazilian State.

5. Average Review Score by Category

A bar chart categorising product categories according to their average customer review scores. This facilitates the identification of high-performing product lines and areas requiring improvement due to diminished satisfaction levels.

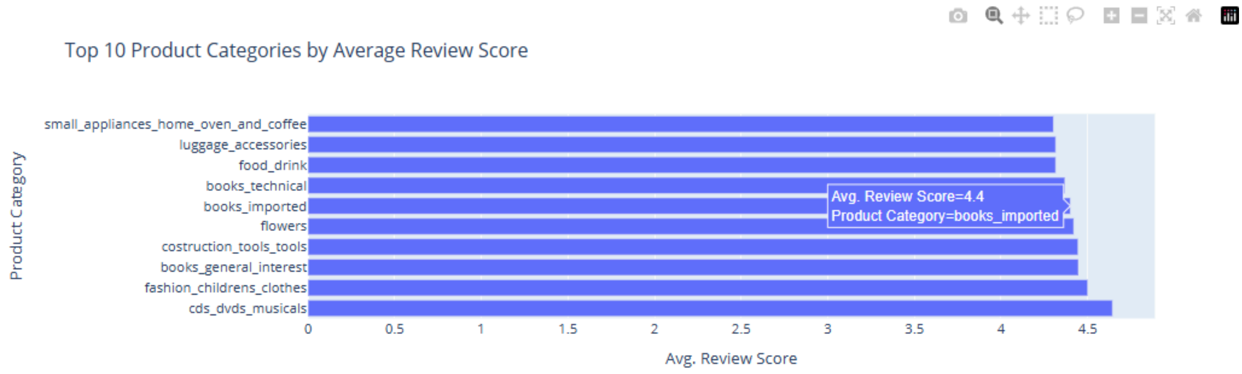

Figure 8 - Average Review Score per Category.

All visualisations were derived from integrated and preprocessed data encompassing orders, customers, products, and reviews. They were selected not only for their analytical significance but also for their capacity to captivate a younger audience through exploration and involvement in a contemporary data landscape.

The display underscores the dataset's business potential and provides a practical basis for implementing ML-driven insights in a real-world retail environment.

5. Data Preparation - Rationale and Justification

A structured and uniform data preparation approach was crucial to guarantee the precision and lucidity of both visualisations and machine learning outcomes. The dataset from Olist comprised several CSV files, each detailing various facets of the e-commerce ecosystem, including orders, items, customers, reviews, and product categories.

The next steps delineate the rationale and justification for each phase of data preparation:

1. Merging Datasets:

The principal datasets were integrated utilising shared identifiers, including *order_id*, *customer_id*, and *product_id*. This facilitated the development of a unified dataset encompassing all pertinent dimensions: customer information, product specifications, purchase history, and review scores.

2. Translating Product Categories

The names of product categories were initially in Portuguese. A translation file was integrated to provide an English version of the category names (product_category_name_english), enhancing readability and accessibility for the intended audience.

3. Datetime Formatting and Feature Extracting

The *order_purchase_timestamp* column was transformed into datetime format to facilitate the extraction of features, including the purchase month. This facilitated temporal analysis and visualisations of trends in consumer behaviour.

4. Filtering Invalid or Incomplete Data

Rows with missing category names or containing non-matching keys were discarded to avoid skewed aggregations and maintain uniform groupings, especially for category-centric analyses.

5. Encoding Transactions for Association Rule Mining

In Market Basket Analysis, transactions were categorised by order_id and product_category_name_english, after that encoded into a binary format to signify the presence or absence of each product category inside each transaction. Transactions containing a single product were omitted to guarantee significant itemset generation.

6. Aggregation for Visualisation

Data for the majority of charts was categorised by relevant dimensions—such as product type, customer state, or month—and subsequently aggregated (count, mean, or total) to generate summary statistics conducive to visual understanding.

7. Category Filtering and Interactivity

In interactive components, such as price distribution by category, only the most pertinent or leading categories were incorporated to enhance responsiveness and prevent clutter. Dropdown filters were filled with distinct values derived from the cleaned category column.

These steps guaranteed that the visualisations relied on accurate and pertinent data while prioritising performance and user experience. Thorough data preparation established the groundwork for the successful execution of machine learning models, facilitating intelligent recommendations and legitimate correlations.

6. Conclusion

This project utilised data visualisation and machine learning methodologies on a real-world e-commerce dataset to derive significant commercial insights. The project produced three primary outputs through a systematic process of data cleaning, integration, and analysis:

1. A recommendation system utilising collaborative filtering to propose pertinent products to clients based on their purchasing behaviour.

2. A Market Basket Analysis (MBA) applying both Apriori and FP-Growth algorithms to identify relationships among product categories.

3. An interactive dashboard targeted for young adults (ages 18–35) that enables users to analyse trends in sales, pricing, geography, and customer happiness.

The collaborative filtering models revealed that data-driven personalisation can improve user engagement, while the MBA uncovered significant cross-selling prospects through strong product connections. The dashboard provided analytical and strategic capabilities, using a simple interface for examining essential indicators and trends.

In combination, these results underscore the collaboration between machine learning and visualisation in enhancing decision-making inside online shopping. The project validated the dataset's quality and richness, demonstrating its suitability for actual business purposes.

7. Business Recommendations

Based on the findings from the recommendation system, Market Basket Analysis, and dashboard insights, the following recommendations are suggested for application in a practical e-commerce setting:

1. Introduce cross-selling mechanisms

Encourage the purchase of frequently co-acquired items (e.g., audio devices with watches and presents) during checkout or on product pages to enhance basket size.

2. Deploy personalised recommendations

Employ item-item collaborative filtering to present personalised product recommendations that mirror genuine client behaviour.

3. Improve product categories with low review scores

Examine categories exhibiting reduced average satisfaction and explore possible factors, including delivery complications or product quality deficiencies.

4. Leverage regional insights for marketing

Utilise geographic data to customise advertising and logistics by state, concentrating efforts on high-priority locations.

5. Price sensitive segmentation

Examine pricing distributions to provide discounts or premium enhancements based on category and target population.

By using these data-driven techniques, online merchants may augment customer experience, optimise operational efficiency, and eventually elevate revenue.

7. References

Provost, F. and Fawcett, T., 2013. *Data Science for Business: What you need to know about data mining and data-analytic thinking.* O'Reilly Media, Inc.

Ricci, F., Rokach, L. and Shapira, B., 2015. *Recommender Systems Handbook.* Springer.

Tan, P.N., Steinbach, M. and Kumar, V., 2005. *Introduction to Data Mining*. Pearson Education.

https://github.com/CCT-Dublin/integrated-ca2-50-FelipeCosta94